

Pharmaniaga Methyldopa Tablet

DESCRIPTION

Yellow, round, biconvex, film-coated tablet with Pharmaniaga icon on one side and scoreline on the other.

COMPOSITION

Each film-coated tablet contains Methyldopa 250 mg.

PHARMACOLOGICAL PROPERTIES

PHARMACODYNAMIC:

Pharmacotherapeutic group: antiadrenergic agents; ATC code C02AB

Mechanism of action

Several mechanisms of action account for the clinically useful effects of methyldopa and the current generally accepted view is that its principal action is on the central nervous system. The antihypertensive effect of methyldopa is probably due to its metabolism to alpha-methylnoradrenaline, which lowers arterial pressure by stimulation of central inhibitory alpha-adrenergic receptors, false neurotransmission, and/or reduction of plasma renin activity.

Methyldopa has been shown to cause a net reduction in the issue concentration of serotonin, dopamine, epinephrine (adrenaline) and norepinephrine (noradrenaline).

PHARMACOKINETIC:

Absorption

Absorption of oral methyldopa is variable and incomplete.

Distribution

Bioavailability after oral administration averages 25%.

Biotransformation

Peak concentrations in plasma occur at two to three hours, and elimination of the drug is biphasic regardless of the route of administration. Plasma half-life is 1.8 ± 0.2 hours.

Elimination

Renal excretion accounts for about two thirds of drug clearance from plasma.

INDICATIONS

Hypertension

CONTRAINDICATIONS

Methyldopa is contraindicated in patients:

- with active hepatic disease, such as acute hepatitis and active cirrhosis
- with liver disorders previously associated with methyldopa therapy
- with hypersensitivity to any component of these products
- on therapy with monoamine oxidase (MAO) inhibitors.

ADVERSE REACTIONS

Sedation, usually transient, may occur during the initial period of therapy or whenever the dose is increased. Headache, asthenia, or weakness may be noted as early and transient symptoms.

The following adverse reactions have been reported and, within each category, are listed in order of decreasing severity.

Cardiovascular: Aggravation of angina pectoris, congestive heart failure, prolonged carotid sinus, hypersensitivity, orthostatic hypotension (decrease daily dosage), edema or weight gain, bradycardia.

Digestive: Pancreatitis, colitis, vomiting, diarrhea, sialadenitis, sore or “black” tongue, nausea, constipation, distension, flatus, dryness of mouth. Endocrine: Hyperprolactinemia.

Hematologic: Bone marrow depression, leukopenia, granulocytopenia, thrombocytopenia, hemolytic anemia; positive tests for antinuclear antibody, LE cells, and rheumatoid factor, positive Coombs test.

Hepatic: Liver disorders including hepatitis, jaundice, abnormal liver function tests.

Hypersensitivity: Myocarditis, pericarditis, vasculitis, lupus-like syndrome, drug-related fever, eosinophilia.

Nervous System/Psychiatric: Parkinsonism, Bell's palsy, decreased mental acuity, involuntary choreoathetotic movements, symptoms of cerebrovascular insufficiency, psychic disturbances including nightmares and reversible mild psychoses or depression, headache, sedation, asthenia or weakness, dizziness, lightheadedness, paresthesias.

Metabolic: Rise in Blood Urea Nitrogen (BUN).

Musculoskeletal: Arthralgia, with or without joint swelling; myalgia.

Respiratory: Nasal stuffiness.

Skin: Toxic epidermal necrolysis, rash.

Urogenital: Amenorrhea, breast enlargement, gynecomastia, lactation, impotence, decreased libido.

WARNINGS AND PRECAUTIONS

It is important to recognize that a positive Coombs test, hemolytic anaemia and liver disorders may occur with methyldopa therapy. The rare occurrences of hemolytic anaemia or liver disorders could lead to potentially fatal complications unless properly recognized and managed. With prolonged methyldopa therapy, 10 to 20 percent of patients develop a positive direct Coombs test which usually occurs between 6 and 12 months of methyldopa therapy. If a positive Coombs test develops during methyldopa therapy, the physician should determine whether hemolytic anaemia exists and whether the positive Coombs test may be a problem.

Before treatment is started, it is desirable to do a blood count (hematocrit, hemoglobin or red cell count) for a baseline or to establish whether there is anaemia. Periodic blood counts should be done during therapy to detect hemolytic anaemia. It may be useful to do a direct Coombs test before therapy and at 6 and 12 months after the start of therapy.

If Coombs-positive hemolytic anaemia occurs, the cause may be methyldopa and the drug should be discontinued. Usually, the anemia remits promptly. If not, corticosteroids may be given and other causes of anemia should be considered. If the hemolytic anaemia is related to methyldopa, the drug should not be reinstituted.

When methyldopa causes Coombs positivity alone or with hemolytic anemia, the red cell is usually coated with gamma globulin of the IgG (gamma G) class only. The positive Coombs test may not revert to normal until weeks to months after methyldopa stopped.

Should the need for transfusion arise in a patient receiving methyldopa, both a direct and an indirect Coombs test should be performed. In the absence of hemolytic anaemia, usually only the direct Coombs test will be positive. A positive direct Coombs test alone will not interfere with typing or cross matching. If the indirect Coombs test is also positive, problems may arise in the major cross match and the assistance of a hematologist or transfusion expert will be needed.

Occasionally, fever has occurred within the first 3 weeks of methyldopa therapy. Jaundice, with or without fever, may occur with onset usually within the first 2 to 3 months of therapy. Rarely, fatal hepatic necrosis has been reported after use of methyldopa. These hepatic changes may represent hypersensitivity reactions. Periodic determinations of hepatic function should be done particularly during the first 6 to 12 weeks of therapy or whenever an unexplained fever occurs. If fever, abnormalities in liver function tests or jaundice appear, stop therapy with methyldopa. If caused by methyldopa, the temperature and abnormalities in liver function characteristically have reverted to normal when the drug was discontinued. Methyldopa should not be reinstituted in such patients.

Rarely, a reversible reduction of the white blood cell count with a primary effect on the granulocytes has been seen. The granulocyte count returned promptly to normal on discontinuance of the drug.

General

Methyldopa should be used with caution in patients with a history of previous liver disease or dysfunction.

Some patients taking methyldopa experience clinical oedema or weight gain which may be controlled by use of a diuretic. Methyldopa should not be continued if oedema progresses or signs of heart failure appear.

Hypertension has recurred occasionally after dialysis in patients given methyldopa because the drug is removed by this procedure.

Rare involuntary choreoathetotic movements have been observed during therapy with methyldopa in patients with severe bilateral cerebrovascular disease. Should these movements occur, stop therapy.

Laboratory Test
Blood count, Coombs test and liver function tests are recommended before initiating therapy and at periodic intervals.

Paediatric Use
There are no well-controlled clinical trials in paediatric patients. Information on dosing in paediatric patients is supported by evidence from published literature regarding the treatment of hypertension in paediatric patients.

Older people use
This drug is known to be substantially excreted by the kidney and the risk of toxic reactions to this drug may be greater in patients with impaired renal function. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection and it may be useful to monitor renal function.

DRUG INTERACTIONS

When methyldopa is used with other antihypertensive drugs, potentiation of antihypertensive effect may occur. Patients should be followed carefully to detect side reactions or unusual manifestations of drug idiosyncrasy.

Patients may require reduced doses of anesthetics when on methyldopa. If hypotension does occur during anesthesia, it usually can be controlled by vasopressors. The adrenergic receptors remain sensitive during treatment with methyldopa.

When methyldopa and lithium are given concomitantly the patient should be carefully monitored for symptoms of lithium toxicity.

PREGNANCY AND LACTATION

Pregnancy
There are no adequate and well-controlled studies in pregnant women in the first trimester of pregnancy.

Breast-feeding
Methyldopa appears in breast milk. Therefore, caution should be exercised when methyldopa is given to a nursing woman.

DOSAGE AND ADMINISTRATION

It should be individualised

Adults:
Initial dosage: Initial dosage is 250mg two/three times a day for 48 hours. Daily dosage may be increased/decreased, preferably at intervals of not less than two days, until an adequate response is achieved. To minimize sedation, start dosage increases in the evening. By adjustment of dosage, morning hypertension may be prevented without sacrificing control of afternoon blood pressure.

Adjustment: The usual daily dosage of Methyldopa is 500mg to 2g in two/four doses. Although occasional patients have responded to higher doses, the maximum recommended daily dosage is 3g. Once an effective

dosage range is attained, a smooth blood pressure response occurs in most patients in 12-24 hours. Since Methyldopa has relatively short duration of action, withdrawal is followed by return of hypertension usually within 48 hours. This is not complicated by an overshoot of blood pressure.

Occasionally tolerance may occur, usually between the second and third month of therapy. Adding a diuretic or increasing the dosage of methyldopa frequently will restore effective control of blood pressure. A thiazide may be added at any time during methyldopa therapy and is recommended if therapy has not been started with a thiazide or if effective control of blood pressure cannot be maintained on 2g of methyldopa daily.

Paediatric population:
Initial dosage is based on 10mg/kg of body weight daily in two to four doses. The daily dosage then is increased or decreased until an adequate response is achieved. The maximum dosage is 65mg/kg or 3g daily, whichever is less.

OVERDOSAGE

Acute overdosage may produce acute hypotension with other responses attributable to brain and gastrointestinal malfunction (excessive sedation, weakness, bradycardia, dizziness, lightheadedness, constipation, distention, flatus, diarrhea, nausea, vomiting).

In the event of overdosage, symptomatic and supportive measures should be employed. When ingestion is recent, gastric lavage or emesis may reduce absorption. When ingestion has been earlier, infusions may be helpful to promote urinary excretion. Otherwise, management includes special attention to cardiac rate and output, blood volume, electrolyte balance, paralytic ileus, urinary function and cerebral activity.

Sympathomimetic drugs [e.g., levarterenol, epinephrine, ARAMINE (Metaraminol Bitartrate)] may be indicated. Methyldopa is dialyzable.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Methyldopa may cause sedation, usually transient, during the initial period of therapy or whenever the dose is increased. If affected, patients should not carry out activities where alertness is necessary, such as driving a car or operating machinery.

ROUTE OF ADMINISTRATION

Oral

STORAGE CONDITIONS

Store below 30°C
Protect from light

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

In blisters of 10, 20, 30, 50, 90, 100, 250, 500 and 1000 tablets.

DATE OF REVISION

28th March 2023

PRODUCT REGISTRATION HOLDER/ MANUFACTURER:
PHARMANIAGA MANUFACTURING BERHAD (198001006232)

No. 11A, Jalan P/1, Kawasan Perusahaan Bangi,
43650 Bandar Baru Bangi, Selangor Darul Ehsan, Malaysia.

PRP 0044.17 280323